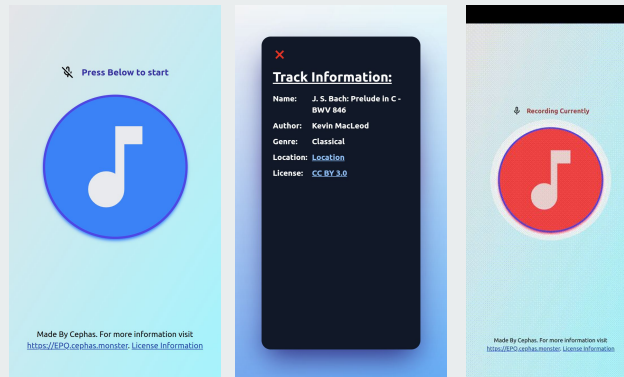



To explore the mechanism that allows music recognition apps to work.

By Cephas




QR codes for all the links

All the links



The artefact





Question

How well can you recognise a song?

I am certainly not good at it. 😞

But how can computers recognise songs better than us?



Why did I choose this title

To explore the mechanism that allows music recognition apps to work.

- Interesting
- It has purpose
- Learning new things
- Difficult / Challenging
- Problem-solving
- Rewarding

Source and resources I used

- Patent from the original creator of Shazam
- Research article
 - How is it made
 - Research about the testing method
- Blog post made by a professional
- Thanks to my computer science teacher listening for feedback.

An oversimplified summary of how it works

The user records the song

🔊 Press Below to start



Made By Cephas. For more information visit <https://github.com/cephas-montesher-License-Information>

The song is uploaded to the server

📶 Recording Currently



Made By Cephas. For more information visit <https://github.com/cephas-montesher-License-Information>

The song is processed

⌚ Please wait...



Made By Cephas. For more information visit <https://github.com/cephas-montesher-License-Information>

The key feature of the song are picked out



An oversimplified summary of how it works

pick out key
feature of the
song

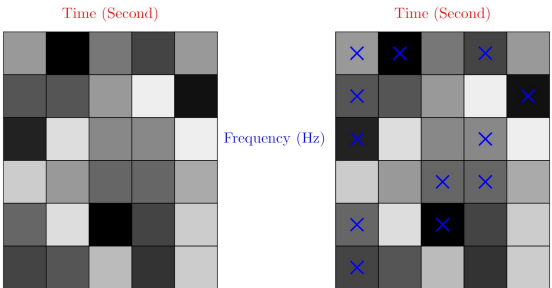
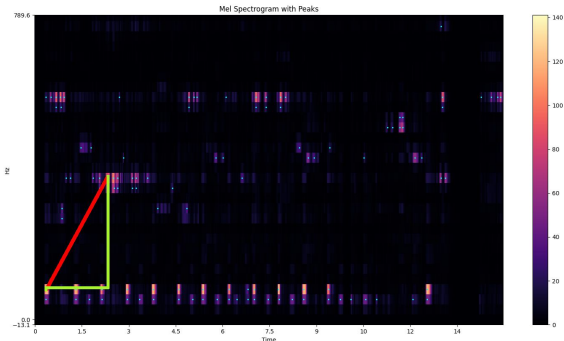


Figure A.4: Illustrating an example of a spectrogram picking of points. The darker the box, the higher the amplitude (value) represents



An oversimplified summary of how it works

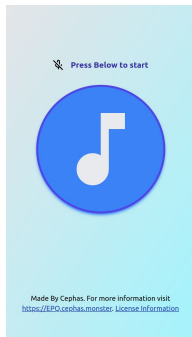
pick out key
feature of the
song



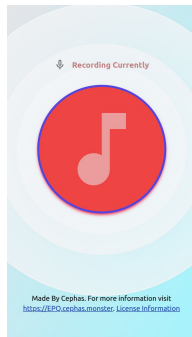
Point A (Hz)	Point B (Hz)	Time delta (s)	Point A time (s)	Track UUID
20.1	302.1	1.8	0.3	ea47a3c5

An oversimplified summary of how it works

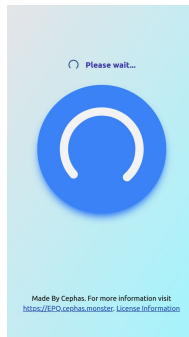
The user records the song



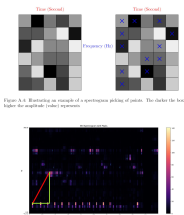
The song are uploaded to the server



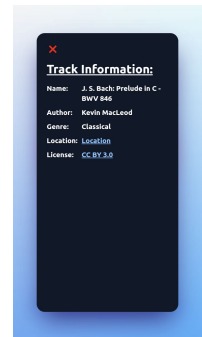
The song are processed



The key feature of the song are picked out



The features are compared with the database



Recording Currently



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Findings

Knowledge and skills:

- What is “Fourier transform”?
- How to make a spectrogram
- How to create the app



Findings

extra programming skills:

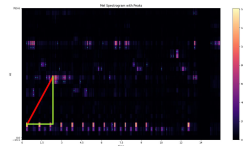
- How to make a Histogram in Python
- How to create an efficient database
- Some data analysis skills

Findings

Plan:

- Testing the effectiveness of the artefact

How can it be improved



- Using “humming” to guess songs doesn’t work.
 - Improving it requires restructuring of the program.
- The music library is too small
 - This project only utilised copyright free songs
 - Just add more
- The program is too slow
 - Money to infrastructure 🚧💡💰



Most importantly

Privacy



Link to the
Privacy notice



Key takeaways

- Time management
- Learning new skills and knowledge quickly
- Problem-solving skills.



Please allow the microphone permission.



Sorry, we unable recognise
this song, because the
duration is too short.



Thank you for listening

Any questions?

All the links



The artefact

